

# THETA–HEISENBERG UNTWISTING, UNIFORM PRILL COVERS, AND ABELIAN BRILL–NOETHER BLOCKS

OPENAI

ABSTRACT. For integers  $g \geq 2$  and  $n \geq 2$  with  $n \mid g$ , I construct a fixed finite unitary representation

$$\rho_{g,n}: \pi_1(\Sigma_g) \longrightarrow SU(n^g)$$

such that, for every marked complex structure  $C$  on  $\Sigma_g$  and every  $L \in \text{Pic}^{g/n}(C)$ , the associated flat bundle satisfies  $H^0(C, \mathcal{V}_{\rho_{g,n}, C} \otimes L) \neq 0$ . The construction identifies the projective Narasimhan–Seshadri monodromy of a Raynaud bundle with the Schrödinger representation of a finite theta group.

I then prove a theta–Heisenberg untwisting theorem. The rank- $n^g$  bundle is realized as the direct image of an exact-order- $n$  torsion line bundle from a degree- $n^g$  cover; a cyclic cover removes this torsion and produces a connected ordinary cover of degree  $n^{g+1}$ ; its Galois closure has degree  $n^{2g+1}$ . These three degrees are sharp among covers retaining the fixed Schrödinger block. Taking  $n = g$  gives, on every curve of genus  $g$ , a connected étale cover of degree  $g^{g+1}$  for which every fibre moves in a pencil. In genus two the degree is 8.

To quantify the remaining gap toward global minimality, I prove a strict rank–corank inequality: a nontrivial stable degree-zero bundle  $W$ , with

$$c = \min_p h^0(W(p)) > 0,$$

satisfies  $\text{rk } W \geq gc + 2$ . This gives the lower bounds  $\deg f \geq gk + 3$  for covers with at least  $k + 1$  fibre sections and  $|\text{Gal}(f)| \geq (g + 2)(g + 3)$  in the Galois Prill-exceptional case. In genus two I determine the minimum over all genus-two curves. On the curve  $y^2 = x^5 - 1$ , a  $C_5$ -equivariant local system with natural  $A_5$ -monodromy has  $w_2 = 1$ , while an equivariant Lefschetz calculation forces its two-dimensional theta obstruction to vanish. It therefore produces a connected Prill-exceptional cover of degree five; the strict rank bound shows that degree five is minimal. The same method extends to Fourier–Mukai bundles pulled back from arbitrary abelian quotients of a Jacobian, producing positive-dimensional abelian blocks in classical Brill–Noether loci.

## 1. INTRODUCTION

Let  $C$  be a smooth projective complex curve and let  $f: X \rightarrow C$  be a connected finite étale cover. The fibre over  $p \in C$  moves in a pencil precisely when

$$h^0(X, \mathcal{O}_X(f^{-1}(p))) \geq 2.$$

Because this dimension is upper semicontinuous in  $p$ , its generic value is its minimum. Thus the condition for a general fibre is equivalent, in this setting, to the same inequality for every fibre. Prill’s problem asks whether this can occur for a general fibre [1, Chapter VI, Exercise D, p. 268]. A degree-36 construction on every genus-two curve was obtained by Landesman and Litt [7]. The higher-genus question was subsequently recorded together with conjectures on generic global generation and horizontal generic vanishing in [9, §6.3].

I prove that every fibre can move, simultaneously for every marked complex structure and in every genus  $g \geq 2$ . The construction is explicit and topological. It does not arise from a single

---

2020 *Mathematics Subject Classification*. 14H30, 14H40, 14K25, 14D20, 20C15, 57K20.

*Key words and phrases*. Raynaud bundle, finite Heisenberg group, theta group, Prill’s problem, Brill–Noether locus, Putman–Wieland conjecture.

isolated algebraic cover, but from a fixed finite local system whose cohomological support locus is the whole Picard variety.

**Theorem 1.1** (Uniform Heisenberg–Raynaud local system). *Let  $g \geq 2$ , let  $n \geq 2$  divide  $g$ , and put*

$$m = \frac{g}{n}, \quad r = n^g.$$

*There is a fixed irreducible representation*

$$\rho_{g,n}: \pi_1(\Sigma_g) \longrightarrow SU(r)$$

*with image of order  $n^{2g+1}$  such that, for every marked genus- $g$  curve  $C$ ,*

$$H^0(C, \mathcal{V}_{\rho_{g,n}, C} \otimes L) \neq 0 \quad \text{for every } L \in \text{Pic}^m(C). \quad (1.1)$$

*The bundle  $\mathcal{V}_{\rho_{g,n}, C}$  is stable of degree zero, has trivial determinant, and has no global sections.*

The divisibility  $n \mid g$  has two simultaneous meanings. It makes the Raynaud slope  $g/n$  integral, so that the Raynaud bundle can be normalized to degree zero by a line bundle. It also kills the scalar obstruction to lifting the theta-group projective representation across the surface relation. Thus the numerical slope and the topological surface obstruction are the same integrality condition.

Lagrangian induction then realizes the high-rank cohomological witness as the direct image of a line bundle on a smaller cover; a cyclic torsor removes its residual scalar monodromy.

**Theorem 1.2** (Theta–Heisenberg untwisting). *Under the hypotheses of [theorem 1.1](#), there is a factorization*

$$X \xrightarrow{v} Y \xrightarrow{u} C$$

*with the following properties.*

- (i) *The cover  $u$  is connected Galois of degree  $r = n^g$ . There is an exact-order- $n$  torsion line bundle  $\lambda \in \text{Pic}^0(Y)$  such that*

$$u_* \lambda \simeq \mathcal{V}_{\rho_{g,n}, C}, \quad H^i(Y, \lambda \otimes u^* L) \simeq H^i(C, \mathcal{V}_{\rho_{g,n}, C} \otimes L) \quad (1.2)$$

*for every line bundle  $L$  and  $i = 0, 1$ .*

- (ii) *The map  $v$  is the connected cyclic degree- $n$  cover trivializing  $\lambda$ . Thus  $f = u \circ v$  has degree  $n^{g+1}$ , and*

$$h^0(X, f^* L) \geq h^0(C, L) + h^0(C, \mathcal{V}_{\rho_{g,n}, C} \otimes L). \quad (1.3)$$

- (iii) *The Galois closure of  $f$  has degree  $n^{2g+1}$ .*

*The degrees  $r$ ,  $nr$ , and  $nr^2$  are respectively minimal among connected twisted-line, ordinary, and Galois covers which retain the fixed Schrödinger local system.*

For  $n = g$ , equation (1.3) applied to  $L = \mathcal{O}_C(p)$  gives the following extension of the genus-two result to every genus.

**Corollary 1.3** (Uniform Prill cover). *For every  $g \geq 2$  there is a fixed finite-index subgroup of  $\pi_1(\Sigma_g)$  such that, for every marked complex structure  $C$ , the induced connected étale cover*

$$f_{g,C}: X_{g,C} \longrightarrow C, \quad \deg f_{g,C} = g^{g+1},$$

*satisfies*

$$h^0(X_{g,C}, \mathcal{O}_{X_{g,C}}(f_{g,C}^{-1}(p))) \geq 2 \quad \text{for every } p \in C$$

*The topological cover is independent of  $C$ . It is non-Galois, and its Galois closure has degree  $g^{2g+1}$ .*

The number  $g^{g+1}$  is not asserted to be the minimum among all Prill-exceptional covers. It is the exact minimum among transitive covers whose permutation local system contains the particular Raynaud–Schrödinger block in [theorem 1.1](#). The distinction is essential. I prove the sharper multiplicity law

$$\deg f = n^{g+1} \operatorname{mult}_{\mathcal{V}_{\rho_{g,n}}}(f_*\mathcal{O}_X) \quad (1.4)$$

for covers obtained from transitive sets of the finite Heisenberg group for which the displayed multiplicity is positive, and the corresponding inequality for arbitrary connected covers retaining the block.

The construction has Brill–Noether consequences. If  $d = n^{g+1}$ , then

$$f^*\operatorname{Pic}^m(C) \subset W_{dm}^0(X), \quad f^*C^{(m)} \subset W_{dm}^1(X).$$

For  $n \geq 3$ , the second locus has Brill–Noether number

$$\rho(g(X), 1, dm) = d(2m - g + 1) - 3 < 0,$$

although it contains an  $m$ -dimensional subvariety. More generally, intermediate isotropic quotients give an exact degree–multiplicity trade-off and higher-rank Brill–Noether strata.

I also obtain a lower bound that does not use finite monodromy.

**Theorem 1.4** (Strict rank–corank inequality). *Let  $W$  be a nontrivial stable vector bundle of degree zero and rank  $r$  on a curve of genus  $g \geq 2$ . If*

$$c = \min_{p \in C} h^0(C, W(p)) \geq 1,$$

*then*

$$r \geq gc + 2. \quad (1.5)$$

Consequently, if every fibre of a connected étale cover has at least  $k + 1$  sections, then its degree is at least  $gk + 3$ . A Galois Prill-exceptional cover has group order at least  $(g + 2)(g + 3)$ . Together with the implication that a Putman–Wieland counterexample is Prill-exceptional [[6](#), Lemma 5.5], this excludes closed Putman–Wieland counterexamples in these numerical ranges. It does not give the converse implication and does not settle the Putman–Wieland conjecture outside those ranges.

A later section extends the method away from the full Jacobian. An ample line bundle on an abelian quotient of a Jacobian supplies a stable semihomogeneous bundle, a Lagrangian line descent, and a finite projective local system. When its slope is integral, Heisenberg untwisting produces a classical rank-one Brill–Noether block on a finite cover. Applied to a Prym subvariety, this gives a Prym-direction support block; it does not by itself give a finite orbit in Prym cohomology.

## 2. FINITE HEISENBERG GROUPS AND SCALAR LIFTING

**2.1. The standard Schrödinger representation.** Fix a primitive  $n$ -th root of unity  $\zeta$ . Define

$$H_{g,n} = \left\langle z, x_1, \dots, x_g, y_1, \dots, y_g \mid \begin{array}{l} z \text{ central, } z^n = x_i^n = y_i^n = 1, \\ [x_i, x_j] = [y_i, y_j] = 1, \quad [x_i, y_j] = z^{\delta_{ij}} \end{array} \right\rangle, \quad (2.1)$$

where  $[a, b] = aba^{-1}b^{-1}$ . The relations collect every word into the following normal form; uniqueness follows from the faithful shift–clock action defined below. Thus every element has a unique expression

$$z^c x_1^{u_1} \cdots x_g^{u_g} y_1^{v_1} \cdots y_g^{v_g},$$

so

$$|H_{g,n}| = n^{2g+1}, \quad Z(H_{g,n}) = \langle z \rangle. \quad (2.2)$$

Let  $T = (\mathbf{Z}/n)^g$ , let  $S = \mathbf{C}[T]$ , and write  $e_t$  for its standard basis. I use the central-weight convention dictated by the dual theta representation in [equation \(3.3\)](#) and define

$$x_i e_t = e_{t+e_i}, \quad y_i e_t = \zeta^{t_i} e_t, \quad z e_t = \zeta^{-1} e_t. \quad (2.3)$$

These matrices satisfy [\(2.1\)](#). The resulting representation

$$\tau_{g,n}: H_{g,n} \longrightarrow U(S)$$

is faithful and irreducible, and  $\dim S = r = n^g$ . I call  $S$  the geometric Schrödinger module. Its dual is the usual weight-one module; fixing the convention here removes any later  $S/S^\vee$  ambiguity.

**Lemma 2.1** (Character and determinant). *The character of  $S$  is*

$$\chi_S(z^a) = r\zeta^{-a}, \quad \chi_S(h) = 0 \quad (h \notin \langle z \rangle). \quad (2.4)$$

If  $g \geq 2$ , then  $\tau_{g,n}(H_{g,n}) \subset SU(S)$ .

*Proof.* If the shift part of  $h$  is nonzero, its matrix has no diagonal entries. If the shift part vanishes but the clock part is nonzero, its trace is a product containing a complete nontrivial geometric sum. This proves [\(2.4\)](#), while irreducibility follows at once from the simultaneous clock eigenspaces and the transitive action of the shifts.

Each  $x_i$  is a product of  $n^{g-1}$  cycles of length  $n$ , and the eigenvalues of  $y_i$  contain every  $n$ -th root with multiplicity  $n^{g-1}$ . Hence

$$\det x_i = \det y_i = (-1)^{(n-1)n^{g-1}} = 1.$$

For even  $n$ , the last equality uses  $g \geq 2$ ; for odd  $n$  it is immediate. Finally,  $\det(\zeta^{-1}I) = \zeta^{-n^g} = 1$ .  $\square$

Set

$$A_{g,n} = H_{g,n}/\langle z \rangle \simeq (\mathbf{Z}/n)^{2g}. \quad (2.5)$$

The scalar commutator in the geometric Schrödinger module induces a perfect alternating pairing

$$e_n: A_{g,n} \times A_{g,n} \longrightarrow \mu_n, \quad e_n(\bar{x}_i, \bar{y}_j) = \zeta^{-\delta_{ij}}. \quad (2.6)$$

It is the inverse of the conventional Weil pairing attached to the dual section module. The finite Stone–von Neumann theorem says that the fixed-central-weight irreducible projective representation is determined, up to projective equivalence, by this pairing [\[11, 15\]](#).

**2.2. Sharpness and the degree–multiplicity law.** For the remainder of this subsection, assume  $n \mid g$ , put  $m = g/n$ , write

$$\pi_g = \left\langle a_1, b_1, \dots, a_g, b_g \left| \prod_{i=1}^g [a_i, b_i] = 1 \right. \right\rangle,$$

and use the surface homomorphism

$$\varphi_{g,n}: \pi_g \longrightarrow H_{g,n}, \quad a_i \longmapsto x_i, \quad b_i \longmapsto y_i.$$

The relation holds because its image is  $z^g = 1$ . Let  $\rho_{g,n} = \tau_{g,n} \circ \varphi_{g,n}$ , and let  $\mathcal{V}_{\rho_{g,n}, C}$  denote its flat bundle on a marked curve. The theta-theoretic support property of this bundle is proved in [theorem 3.3](#); the degree statements below are purely representation-theoretic.

The next theorem makes the minimality statement precise without making a claim about unrelated Prill constructions.

**Theorem 2.2** (Three sharp degree bounds). *Let  $V = \mathcal{V}_{\rho_{g,n}, C}$ , let  $r = n^g$ , and let  $\varphi = \varphi_{g,n}$ .*

- (i) Suppose  $u' : Y' \rightarrow C$  is connected and  $\eta$  is a rank-one finite local system on  $Y'$ . If  $V$  is a summand of  $u'_*\eta$ , then  $\deg u' \geq r$ .
- (ii) Suppose  $f' : X' \rightarrow C$  is connected and  $V$  occurs in  $f'_*\mathcal{O}_{X'}$  with multiplicity  $q > 0$ . Then

$$\deg f' \geq nrq = n^{g+1}q. \quad (2.7)$$

If  $f'$  is the cover associated with a transitive  $H_{g,n}$ -set, equality holds:

$$\deg f' = nrq. \quad (2.8)$$

- (iii) Suppose  $f'$  is connected Galois and  $V$  is a summand of  $f'_*\mathcal{O}_{X'}$ . Then  $\deg f' \geq nr^2 = n^{2g+1}$ .

The covers in [theorem 4.1](#) attain the three respective lower bounds.

*Proof.* Let  $\Gamma \subset \pi_g$  be the subgroup defining a connected cover and put  $J = \varphi(\Gamma) \subset H = H_{g,n}$ . The map of coset sets  $\pi_g/\Gamma \rightarrow H/J$  is surjective, so

$$[\pi_g : \Gamma] \geq [H : J]. \quad (2.9)$$

For (i), Frobenius reciprocity gives a one-dimensional character of  $\Gamma$  occurring in  $S|_\Gamma$ . Its restriction to  $\Gamma \cap \ker \varphi$  is trivial because that subgroup acts as the identity on  $S$ ; hence it descends to a character of  $J$ . Commutators in  $J$  act trivially on this line. The faithful central character of  $S$  therefore shows that the image of  $J$  in  $A_{g,n}$  is isotropic. A subgroup of a finite symplectic module of order  $r^2$  has order at most  $r$ . Writing  $\bar{J}$  for that image gives

$$|J| = |J \cap \langle z \rangle| |\bar{J}| \leq nr.$$

Thus  $[H : J] \geq r$ .

For (ii), Frobenius reciprocity gives

$$q = \dim(S^\vee)^J. \quad (2.10)$$

This dimension is positive only if  $J \cap \langle z \rangle = 1$ . Using [lemma 2.1](#),

$$q = \frac{1}{|J|} \sum_{j \in J} \chi_{S^\vee}(j) = \frac{r}{|J|}. \quad (2.11)$$

Consequently,

$$[H : J] = \frac{nr^2}{|J|} = nrq.$$

Together with (2.9), this proves (2.7); equality holds when  $\Gamma = \varphi^{-1}(J)$ .

For (iii),  $\Gamma$  is normal, so  $J$  is normal in  $H$ . The nonzero invariant space  $(S^\vee)^J$  is then  $H$ -stable. Irreducibility gives  $(S^\vee)^J = S^\vee$ , and faithfulness gives  $J = 1$ . Equation (2.9) now yields  $\deg f' \geq |H| = nr^2$ .  $\square$

*Remark 2.3* (Why the projective orbit is too small). The stabilizer  $B = \langle z \rangle K$  has index  $r$ , but its ordinary permutation module factors through  $H/\langle z \rangle$ . The centre therefore acts trivially on every constituent, so that permutation module cannot contain  $S$ . The degree- $r$  stage retains  $S$  only after inserting the character line  $\lambda$ . Removing that scalar costs exactly the cyclic factor  $n$ .

**2.3. The isotropic tower.** For  $0 \leq k \leq g$ , set

$$K_k = \langle y_1, \dots, y_k \rangle, \quad \Gamma_k = \varphi_{g,n}^{-1}(K_k),$$

and let  $f_k : X_k \rightarrow C$  be the associated cover. Define

$$d_k = n^{2g+1-k}, \quad \mu_k = n^{g-k}. \quad (2.12)$$

**Proposition 2.4** (Exact compression tower). *The covers  $f_k$  are connected and satisfy*

$$\deg f_k = d_k, \quad \text{mult}_S \mathbf{C}[H_{g,n}/K_k] = \mu_k, \quad d_k = n^{g+1} \mu_k. \quad (2.13)$$

*There is a tower*

$$X_0 \longrightarrow X_1 \longrightarrow \cdots \longrightarrow X_g \longrightarrow C \quad (2.14)$$

*in which every map  $X_{k-1} \rightarrow X_k$  is cyclic étale of degree  $n$ . The cover  $X_0 \rightarrow C$  is Galois with group  $H_{g,n}$ ; every  $X_k \rightarrow C$  for  $k > 0$  is non-Galois and has the same Galois closure. Moreover,*

$$(f_k)_* \mathcal{O}_{X_k} \simeq \mathcal{O}_C \oplus \mathcal{V}_{\rho_{g,n},C}^{\oplus \mu_k} \oplus E_k \quad (2.15)$$

*and, for every  $L \in \text{Pic}^m(C)$ ,*

$$h^0(X_k, f_k^* L) \geq h^0(C, L) + \mu_k. \quad (2.16)$$

*Proof.* The order and multiplicity statements follow from (2.11). The inclusions  $K_{k-1} \triangleleft K_k$  give the cyclic maps in (2.14). Conjugating a nontrivial element of  $K_k$  by the corresponding  $x_i$  produces a nontrivial central factor; hence  $K_k$  is not normal for  $k > 0$ , and the same argument as in theorem 4.1 shows that its core is trivial. The representation splitting and (2.16) follow from Frobenius reciprocity, the projection formula, and theorem 3.3.  $\square$

**Corollary 2.5** (Prill’s problem in every genus). *Taking  $n = g$  and  $k = g$  gives a fixed topological cover whose holomorphic realization  $f_{g,C}: X_{g,C} \rightarrow C$ , for every marked  $C$ , is connected and étale of degree*

$$g^{g+1}$$

*such that every fibre moves in a pencil. It is fixed by the marking and works for every complex structure. Its source has genus*

$$1 + g^{g+1}(g - 1). \quad (2.17)$$

*Proof.* Here  $m = 1$  and  $\mu_g = 1$ . Apply (2.16) to  $L = \mathcal{O}_C(p)$  and use Riemann–Hurwitz.  $\square$

**Example 2.6** (Genus two). For  $g = n = 2$ , put  $K = K_g = \langle y_1, y_2 \rangle$ . The  $K$ -quotient cover, which is minimal among ordinary covers retaining the fixed Schrödinger block, has degree 8 and source genus 9; its Galois closure has degree 32. If  $\alpha_1, \alpha_2, \alpha_3$  are the three nontrivial characters appearing in the annihilator of  $K$ , then

$$f_* \mathcal{O}_X \simeq \mathcal{O}_C \oplus \alpha_1 \oplus \alpha_2 \oplus \alpha_3 \oplus V, \quad (2.18)$$

where  $V$  is the rank-four Schrödinger–Raynaud bundle. The strict rank theorem below forces  $\min_p h^0(V(p)) = 1$ , and this minimum holds on a nonempty open set. For each nontrivial  $\alpha_i$ , the curves  $C$  and  $C - \alpha_i$  in  $J(C)$  have no common component, so  $h^0(\alpha_i(p)) = 0$  for general  $p$ . Hence a general fibre has exactly two sections. This lowers the degree in the every-curve conclusion from 36 to 8, compared with [7]. This construction is uniform over every genus-two curve, but it is not globally minimal. The degree-five example in theorem 6.10 shows that the minimum over all genus-two curves and connected Prill-exceptional covers is five.

The Galois closure gives a complementary sharp count. Since the regular representation of  $H_{2,2}$  decomposes as

$$\tilde{f}_* \mathcal{O}_{\tilde{X}} \simeq \bigoplus_{\eta \in (\mathbf{Z}/2)^4} \alpha_\eta \oplus V^{\oplus 4},$$

the degree-32 cover has source genus 33, and every fibre has at least five sections. On the same minimum-open set used above, all nontrivial  $\alpha_\eta(p)$  have no section, so a general fibre has exactly five.

**2.4. The surface scalar.** Write

$$\pi_g = \left\langle a_1, b_1, \dots, a_g, b_g \left| \prod_{i=1}^g [a_i, b_i] = 1 \right. \right\rangle.$$

**Lemma 2.7** (Surface lifting obstruction). *Let  $A$  be a finite abelian group equipped with a projective representation whose commutator is an alternating pairing  $e: A \times A \rightarrow \mathbf{C}^*$ . For a homomorphism  $\alpha: \pi_g \rightarrow A$ , a choice of lifts of  $\alpha(a_i), \alpha(b_i)$  to the associated central extension satisfies the surface relation if and only if*

$$\kappa(\alpha) := \prod_{i=1}^g e(\alpha(a_i), \alpha(b_i)) = 1. \quad (2.19)$$

The condition is independent of all chosen lifts.

*Proof.* Changing a lift by a central scalar does not change a commutator. The image of the defining surface relation is therefore exactly the product in (2.19). If the product is one, arbitrary generator lifts define a homomorphism; if it is not one, no scalar change can repair the relation.  $\square$

For the mod- $n$  abelianization in a marked symplectic basis,  $\kappa = \zeta^{-g}$ . Thus the standard projective Schrödinger representation has a finite linear surface lift precisely when  $n \mid g$ . Under this assumption define

$$\varphi_{g,n}: \pi_g \twoheadrightarrow H_{g,n}, \quad a_i \mapsto x_i, \quad b_i \mapsto y_i, \quad (2.20)$$

and set

$$\rho_{g,n} = \tau_{g,n} \circ \varphi_{g,n}. \quad (2.21)$$

The map is surjective because the commutators generate the centre.

### 3. RAYNAUD BUNDLES AND PROJECTIVE TRANSPORT

**3.1. The theta-equivariant Fourier–Mukai bundle.** Let  $C$  be a marked smooth projective curve of genus  $g$ , let  $J = \text{Pic}^0(C)$ , and choose an Abel–Jacobi embedding  $\iota: C \hookrightarrow J$ . Choose a symmetric principal theta divisor  $\Theta$ , and put  $N = \mathcal{O}_J(n\Theta)$ . If  $\Phi_{\mathcal{P}}$  denotes the Fourier–Mukai transform for the normalized Poincaré bundle, set

$$F_n = \Phi_{\mathcal{P}}(N^{-1})[g], \quad R_n = \iota^* F_n. \quad (3.1)$$

Raynaud and Schneider prove the following facts [14, 15]:

$$\text{rk } R_n = n^g, \quad \mu(R_n) = \frac{g}{n}, \quad H^0(C, R_n \otimes \alpha) \neq 0 \quad (\alpha \in \text{Pic}^0(C)), \quad (3.2)$$

and  $R_n$  is stable. The key equivariant identity is

$$[n]^* F_n \simeq \mathcal{O}_J(n\Theta) \otimes H^0(J, \mathcal{O}_J(n\Theta))^\vee \quad (3.3)$$

in the following precise equivariant sense; see [15, Proposition 2.1(3)]. Under the principal-polarization identification  $J \simeq \widehat{J}$ , the polarization isogeny of  $\mathcal{O}_J(n\Theta)$  is  $[n]$ . Its theta group  $\mathcal{G}_n = \mathcal{G}(\mathcal{O}_J(n\Theta))$  acts on the right-hand side of (3.3): the centre  $\mathbf{C}^*$  has weight  $+1$  on the line-bundle factor and weight  $-1$  on the dual section factor. The tensor product therefore has central weight zero and carries an honest action of  $\mathcal{G}_n/\mathbf{C}^* = J[n]$ . A theta structure identifies the action on  $H^0(J, \mathcal{O}_J(n\Theta))^\vee$  with the fixed geometric Schrödinger module  $S$  in (2.3). Its commutator is the inverse Weil pairing, as prescribed by the chosen central-weight convention.

Let

$$q: \widetilde{C} = C \times_{J[n]} J \longrightarrow C. \quad (3.4)$$

The cover is connected: its monodromy is the surjective mod- $n$  abelianization  $\pi_1(C) \rightarrow H_1(C, \mathbf{Z}/n)$ . Restricting (3.3) gives

$$q^*R_n \simeq \mathcal{N} \otimes S \quad (3.5)$$

for a line bundle  $\mathcal{N}$  on  $\tilde{C}$ , with the full theta group acting on the descent data.

**Lemma 3.1** (Projective theta descent). *After the marking identifies  $J[n]$  with  $A_{g,n}$ , the projective bundle  $\mathbf{P}(R_n)$  is the bundle associated to the fixed projective representation*

$$\pi_g \longrightarrow A_{g,n} \longrightarrow PGL(S) \quad (3.6)$$

*given by the mod- $n$  abelianization and the projective Schrödinger representation on  $S$ . Its projective conjugacy class is independent of  $C$ , of the Abel–Jacobi base point, and of the chosen theta structure.*

*Proof.* Projectivizing (3.5) removes the line factor  $\mathcal{N}$ , so  $q^*\mathbf{P}(R_n)$  is the constant  $\mathbf{P}(S)$ -bundle. Write  $G = J[n]$  for the deck group. Under the theta-equivariant form of (3.3), the descent isomorphism attached to  $a \in G$  is represented on  $\mathcal{N} \otimes S$  by

$$\ell_a \otimes T_a,$$

where  $\ell_a: a^*\mathcal{N} \rightarrow \mathcal{N}$  is a line isomorphism and  $T_a \in GL(S)$  is a theta-group lift. Replacing a lift by a scalar rescales  $\ell_a$  inversely, so  $[T_a] \in PGL(S)$  is intrinsic. The descent cocycle gives  $[T_{a+b}] = [T_a T_b]$ , while

$$T_a T_b T_a^{-1} T_b^{-1} = e_n(a, b)I.$$

Thus the projective deck action is the constant Schrödinger action with commutator the inverse Weil pairing. The cover  $q$  is the pullback of  $[n]$  along the Abel–Jacobi map, hence its monodromy is the marked mod- $n$  abelianization. The marking and the convention in (2.3) therefore identify the descent representation with (3.6).

Choose invariant Hermitian metrics on the finite theta representation and on the descended line factor. The preceding cocycle then defines a unitary flat connection on  $\mathbf{P}(R_n)$  whose holomorphic structure is the one obtained from theta descent. The stable bundle  $R_n$  has a unique Hermitian–Einstein connection up to unitary scalar gauge, and its projectivization is flat. Uniqueness identifies its projective connection with the descended unitary connection. Thus algebraic theta descent and projective Narasimhan–Seshadri monodromy give the same projective local system.

Changing the theta structure conjugates the projective representation. Changing the theta divisor or the Abel–Jacobi base point translates a semihomogeneous Fourier–Mukai bundle; the translation formula  $t_a^*F_n \simeq F_n \otimes P_b$ , for a suitable  $P_b \in \text{Pic}^0(J)$ , changes its restriction only by a line twist. Consequently the projectivization is unchanged. There is no even-level defect: since  $\mathbf{C}^*$  is divisible,

$$H^2(A_{g,n}, \mathbf{C}^*) \simeq \text{Hom}(\wedge^2 A_{g,n}, \mathbf{C}^*),$$

so the projective cohomology class is determined by its commutator. Quadratic refinements affect extensions of symplectic automorphisms, not the marked projective representation used here.  $\square$

**3.2. Scalar transport of support loci.** For a bundle  $E$  on  $C$ , define

$$V_d^i(E) = \{L \in \text{Pic}^d(C) : H^i(C, E \otimes L) \neq 0\}. \quad (3.7)$$

**Lemma 3.2** (Projective scalar transport). *Let  $E$  and  $E'$  be vector bundles of the same rank on a curve. If  $\mathbf{P}(E) \simeq \mathbf{P}(E')$  over the curve, then*

$$E' \simeq E \otimes \alpha \quad \text{for some line bundle } \alpha. \quad (3.8)$$

If  $\deg E = \deg E'$ , then  $\alpha \in \text{Pic}^0(C)$ . Consequently,

$$V_d^i(E') = \alpha^{-1} V_d^i(E). \quad (3.9)$$

In particular, the property  $V_d^i(E) = \text{Pic}^d(C)$  depends only on the projective local system.

*Proof.* The two bundles define the same  $PGL_r$ -torsor. Comparing lifts through  $1 \rightarrow \mathbf{G}_m \rightarrow GL_r \rightarrow PGL_r \rightarrow 1$  shows that their transition matrices differ by a  $\mathbf{G}_m$ -cocycle, which is a line bundle  $\alpha$ . This proves (3.8). Taking degrees gives  $\deg E' = \deg E + r \deg \alpha$ , so equal degrees force  $\deg \alpha = 0$ . Formula (3.9) follows by tensoring line bundles.  $\square$

**Theorem 3.3** (Heisenberg–Raynaud uniformization). *Let  $g \geq 2$ ,  $n \geq 2$ , and  $n \mid g$ . The representation  $\rho_{g,n}$  in (2.21) satisfies all conclusions of theorem 1.1.*

*Proof.* Put  $m = g/n$  and  $r = n^g$ . Choose  $M \in \text{Pic}^{-m}(C)$  with

$$M^{\otimes r} \simeq (\det R_n)^{-1}; \quad (3.10)$$

such an  $M$  exists because multiplication by  $r$  is surjective on  $\text{Pic}^0(C)$ . The bundle

$$W = R_n \otimes M$$

is stable of degree zero and has trivial determinant. By the Narasimhan–Seshadri theorem [12], it is associated to an irreducible unitary representation  $\sigma_C$ .

The projective local system of  $\sigma_C$  is the one in lemma 3.1. The fixed finite representation  $\rho_{g,n}$  is a linear lift of the same projective representation by lemma 2.7. Therefore lemma 3.2 gives an  $\alpha_C \in \text{Pic}^0(C)$  such that

$$\mathcal{V}_{\rho_{g,n},C} \simeq R_n \otimes M \otimes \alpha_C. \quad (3.11)$$

For  $L \in \text{Pic}^m(C)$ , the line  $M \otimes \alpha_C \otimes L$  has degree zero. Equation (3.2) now proves

$$H^0(C, \mathcal{V}_{\rho_{g,n},C} \otimes L) \neq 0.$$

The representation is irreducible and nontrivial, so its degree-zero flat bundle is stable and has no global sections. Triviality of the determinant follows directly from lemma 2.1. Its image is the faithful Schrödinger image of  $H_{g,n}$ , hence has order  $n^{2g+1}$ .  $\square$

*Remark 3.4* (Logical scope). The uniformity in theorem 3.3 is not a compactness or finite-candidate argument. It uses two exact inputs: theta-equivariant projective descent and invariance of the full Picard support condition under scalar twists. A statement only along the Abel curve  $p \mapsto \mathcal{O}_C(p)$  would not be sufficient for this transport.

#### 4. LAGRANGIAN DESCENT AND HEISENBERG UNTWISTING

**4.1. The three-step factorization.** Put

$$K = \langle y_1, \dots, y_g \rangle, \quad B = \langle z \rangle K. \quad (4.1)$$

Then  $K \simeq (\mathbf{Z}/n)^g$ ,  $B = \langle z \rangle \times K$ , and

$$|K| = r, \quad |B| = nr. \quad (4.2)$$

The subgroup  $B$  is normal in  $H_{g,n}$ , since conjugation changes an element of  $B$  only by a central element. Define

$$\chi: B \longrightarrow \mu_n, \quad \chi(z) = \zeta^{-1}, \quad \chi(y_i) = 1. \quad (4.3)$$

The line  $\mathbf{C}e_0 \subset S$  is a  $B$ -eigenline of character  $\chi$ . It is cyclic under the shifts, and therefore

$$S \simeq \text{Ind}_B^{H_{g,n}} \chi. \quad (4.4)$$

**Theorem 4.1** (Exact untwisting). *Let  $C$  be marked and retain the hypotheses of [theorem 3.3](#). Let*

$$\pi_B = \varphi_{g,n}^{-1}(B), \quad \pi_K = \varphi_{g,n}^{-1}(K).$$

*The corresponding covers form a diagram*

$$X \xrightarrow{v} Y \xrightarrow{u} C \quad (4.5)$$

*with the following properties.*

- (a) *The cover  $u$  is connected Galois of degree  $r$ .*
- (b) *The character  $\chi \circ \varphi_{g,n}|_{\pi_B}$  defines an exact-order- $n$  torsion line bundle  $\lambda$  on  $Y$ , chosen with the associated-bundle convention for which*

$$u_*\lambda \simeq \mathcal{V}_{\rho_{g,n},C}. \quad (4.6)$$

*If the opposite convention is used,  $\lambda$  is replaced by  $\lambda^{-1}$ .*

- (c) *The cover  $v$  is connected cyclic étale of degree  $n$ ,  $v^*\lambda \simeq \mathcal{O}_X$ , and*

$$X \simeq \operatorname{Spec}_Y \left( \bigoplus_{j=0}^{n-1} \lambda^{-j} \right) \quad (4.7)$$

*after a choice of  $\lambda^n \simeq \mathcal{O}_Y$ .*

- (d) *For every line bundle  $L$  on  $C$  and  $i = 0, 1$ ,*

$$H^i(Y, \lambda \otimes u^*L) \simeq H^i(C, \mathcal{V}_{\rho_{g,n},C} \otimes L). \quad (4.8)$$

- (e) *The composite  $f = u \circ v$  has degree  $nr = n^{g+1}$ , and*

$$f_*\mathcal{O}_X \simeq \mathcal{O}_C \oplus \mathcal{V}_{\rho_{g,n},C} \oplus E \quad (4.9)$$

*for a unitary degree-zero bundle  $E$ .*

- (f) *The Galois closure of  $f$  is the  $H_{g,n}$ -cover and has degree  $nr^2 = n^{2g+1}$ .*

*Proof.* Equation (4.4) and the equivalence between finite-cover direct image and induction give (4.6). The character is surjective because its restriction to the centre is faithful; hence  $\lambda$  has exact order  $n$ . Its kernel in  $\pi_B$  is exactly  $\pi_K$ , which proves the assertions about  $v$  and (4.7). Finite direct image is exact, and the projection formula proves (4.8).

The permutation module of the composite is  $\mathbf{C}[H_{g,n}/K] = \operatorname{Ind}_K^{H_{g,n}} \mathbf{1}$ . Frobenius reciprocity and the fixed lines in both  $S$  and  $S^\vee$  show that it contains  $S$  once. It also contains the trivial representation, proving (4.9).

It remains to determine the Galois closure. If  $k$  belongs to the core of  $K$ , then  $[h, k] \in K \cap \langle z \rangle = 1$  for every  $h \in H$ . The nondegeneracy of (2.6) forces the image of  $k$  in  $H/\langle z \rangle$  to vanish, and then  $k = 1$ . Thus the core is trivial.  $\square$

**4.2. Finite-symplectic theta untwisting.** The preceding argument is not restricted to a homogeneous module  $(\mathbf{Z}/n)^{2g}$ . The following form is the group-theoretic engine needed for arbitrary polarization type.

**Theorem 4.2** (Finite-symplectic theta untwisting). *Let  $(A, e)$  be a finite abelian group with a perfect alternating pairing  $e: A \times A \rightarrow \mu_N$ , where  $N$  is the exponent of  $A$ , assume  $N \geq 2$ , and write  $|A| = r^2$ . Choose a Lagrangian decomposition  $A = L \oplus L'$ , and let  $H_{\text{sp}}(A)$  be the split finite Heisenberg model described in the proof. Suppose a surface group admits a surjection  $\psi: \pi_1(C) \twoheadrightarrow H_{\text{sp}}(A)$ . For a faithful central character, let  $S_A$  be the corresponding Schrödinger module and  $V_A$  its flat bundle. Then there are connected covers*

$$X_A \longrightarrow Y_A \longrightarrow C$$

and an exact-order- $N$  line bundle  $\lambda_A \in \text{Pic}^0(Y_A)$  such that

$$\deg(Y_A/C) = r, \quad (u_A)_*\lambda_A \simeq V_A,$$

the first map is the cyclic degree- $N$  torsor trivializing  $\lambda_A$ , and

$$(f_A)_*\mathcal{O}_{X_A} \simeq \mathcal{O}_C \oplus V_A \oplus E_A.$$

Thus  $\deg f_A = Nr$ , and the Galois closure of  $f_A$  is the  $H_{\text{sp}}(A)$ -cover of degree  $Nr^2$ . For  $i = 0, 1$ , finite direct image identifies the cohomology of  $V_A \otimes L$  with that of  $\lambda_A \otimes u_A^*L$ .

*Proof.* The elementary-divisor theorem for finite symplectic modules gives a decomposition

$$A = L \oplus L', \quad |L| = |L'| = r,$$

in which both summands are isotropic and the pairing between them is perfect. Concretely, if  $A \simeq \bigoplus_i (\mathbf{Z}/d_i)^2$ , take one cyclic factor from each hyperbolic plane. The split finite model in the statement is

$$H_{\text{sp}}(A) = \mu_N \times L \times L',$$

with multiplication

$$(z, l, l')(w, m, m') = (zw e(l', m), l + m, l' + m').$$

Its commutator is the prescribed pairing. The projective cocycle class with commutator  $e$  is unique because

$$H^2(A, \mathbf{C}^*) \simeq \text{Hom}(\wedge^2 A, \mathbf{C}^*),$$

and the finite Stone–von Neumann theorem gives the unique irreducible Schrödinger projective representation for each faithful central character. This does not assert that every finite central extension with the same commutator is isomorphic to the split group.

Put  $B = \mu_N \times L$ . For a faithful character  $\omega: \mu_N \rightarrow \mathbf{C}^*$ , extend it to  $\chi(z, l) = \omega(z)$  on  $B$ . Then

$$S_A \simeq \text{Ind}_B^{H_{\text{sp}}(A)} \chi,$$

so  $\dim S_A = [H_{\text{sp}}(A) : B] = r$ . The group  $B$  is normal. The subgroups  $\psi^{-1}(B)$  and  $\psi^{-1}(L)$  therefore define the asserted degree- $r$  cover and its cyclic degree- $N$  refinement. Surjectivity of  $\psi$  and faithfulness of  $\omega$  show that the associated character line has exact order  $N$ . Induction, finite direct image, and the projection formula give the direct-image and cohomology identities.

The permutation module  $\mathbf{C}[H_{\text{sp}}(A)/L]$  contains the trivial module and also  $S_A$ , since the dual Schrödinger module has an  $L$ -fixed inducing line and Frobenius reciprocity applies. Finally, the core of  $L$  is trivial. Indeed, if  $l \in L$  belongs to every conjugate of  $L$ , conjugation by every  $l' \in L'$  forces  $e(l', l) = 1$ ; perfectness gives  $l = 0$ . Hence the Galois closure has group  $H_{\text{sp}}(A)$  and order  $Nr^2$ .  $\square$

**Corollary 4.3** (Higher-divisor Prill condition). *Let  $f = u \circ v: X \rightarrow C$  be the composite in theorem 4.1. Then, for every effective divisor  $D$  of degree  $m = g/n$ ,*

$$h^0(X, \mathcal{O}_X(f^*D)) \geq 2. \tag{4.10}$$

*More strongly, for every  $L \in \text{Pic}^m(C)$ ,*

$$h^0(X, f^*L) \geq h^0(C, L) + 1. \tag{4.11}$$

*Proof.* Apply (4.9), the projection formula, and theorem 3.3. If  $D$  is effective, the trivial summand gives one section and the Schrödinger summand gives another.  $\square$

## 5. BRILL–NOETHER GEOMETRY OF THE COMPRESSION TOWER

Retain  $m = g/n$ , and let  $f_k: X_k \rightarrow C$  be the covers from [proposition 2.4](#). Put

$$G_k = g(X_k) = 1 + d_k(g - 1), \quad e_k = md_k. \quad (5.1)$$

I use the convention

$$W_e^s(X) = \{L \in \text{Pic}^e(X) : h^0(X, L) \geq s + 1\}.$$

**Theorem 5.1** (Abelian and symmetric-product blocks). *For every  $0 \leq k \leq g$ ,*

$$f_k^* \text{Pic}^m(C) \subset W_{e_k}^{\mu_k - 1}(X_k) \quad (5.2)$$

*is a translate of an abelian subvariety of dimension  $g$ . Moreover,*

$$f_k^* W_m^0(C) \subset W_{e_k}^{\mu_k}(X_k) \quad (5.3)$$

*contains the image of an  $m$ -dimensional symmetric product. The corresponding Brill–Noether numbers are*

$$\rho_k = 1 + d_k(g - 1) - \mu_k[d_k(g - 1 - m) + \mu_k] \quad (5.4)$$

*and*

$$\rho_k^+ = 1 + d_k(g - 1) - (\mu_k + 1)[1 + d_k(g - 1 - m) + \mu_k]. \quad (5.5)$$

*Proof.* Equation (2.16) gives  $h^0(X_k, f_k^* L) \geq \mu_k$  for every  $L \in \text{Pic}^m(C)$ , and gives  $h^0(X_k, f_k^* L) \geq \mu_k + 1$  when  $L$  is effective. This proves the two inclusions, where  $C^{(m)} \rightarrow W_m^0(C)$  is the Abel–Jacobi map.

The identity

$$\text{Nm}_{f_k} \circ f_k^* = [d_k]: \text{Pic}^0(C) \longrightarrow \text{Pic}^0(C) \quad (5.6)$$

shows that  $f_k^*$  has finite kernel. Hence the first image has dimension  $g$ . Since  $m \leq g - 1$ , a general effective divisor of degree  $m$  has one section, so  $\dim W_m^0(C) = m$ ; the constructed second image therefore has dimension  $m$ . Substitution of  $G_k = 1 + d_k(g - 1)$ ,  $e_k = md_k$ , and the relevant values of  $s$  in

$$\rho(G, s, e) = G - (s + 1)(G - e + s)$$

gives (5.4) and (5.5).  $\square$

**Corollary 5.2** (Negative expected dimension). *At the minimal ordinary layer  $k = g$ , put  $d = n^{g+1}$  and write  $f = f_g: X = X_g \rightarrow C$ . Then*

$$f^* \text{Pic}^m(C) \subset W_{dm}^0(X), \quad f^* C^{(m)} \subset W_{dm}^1(X). \quad (5.7)$$

*The second locus has*

$$\rho(g(X), 1, dm) = d(2m - g + 1) - 3. \quad (5.8)$$

*If  $n \geq 3$ , this number is negative, although the constructed image  $f^* C^{(m)}$  has dimension  $m$ . In particular, for  $n = g$  and  $g \geq 3$ ,*

$$f^* C \subset W_d^1(X), \quad \rho(g(X), 1, d) = d(3 - g) - 3 < 0. \quad (5.9)$$

*Proof.* Here  $\mu_g = 1$ . Formula (5.8) follows directly from  $g(X) = 1 + d(g - 1)$ . If  $n \geq 3$ , then  $g \geq n$  and

$$2m - g + 1 = \frac{2g}{n} - g + 1 \leq 0;$$

equality occurs only at the boundary  $g = n = 3$ , where the remaining term is  $-3$ .  $\square$

Write

$$d_s(X) = \min\{\deg L : L \in \text{Pic}(X), h^0(X, L) \geq s + 1\}$$

for the gonality sequence.

**Corollary 5.3** (Gonality sequence and Clifford index). *For every effective divisor  $D$  of degree  $m$ ,*

$$h^0(X_k, \mathcal{O}_{X_k}(f_k^* D)) \geq \mu_k + 1, \quad h^1(X_k, \mathcal{O}_{X_k}(f_k^* D)) \geq \mu_k + 1 + d_k(g - 1 - m). \quad (5.10)$$

Consequently,

$$d_{\mu_k}(X_k) \leq e_k, \quad \text{Cliff}(X_k) \leq e_k - 2\mu_k. \quad (5.11)$$

*Proof.* The first inequality is (2.16). Riemann–Roch gives the second. Since  $m \leq g/2$ , both  $h^0$  and  $h^1$  are at least two, and the definitions of the gonality sequence and Clifford index give (5.11).  $\square$

## 6. A STRICT RANK–CORANK INEQUALITY

The argument in this section applies to arbitrary stable degree-zero bundles; neither finite monodromy nor the theta construction is used.

**Theorem 6.1** (Strict rank–corank inequality). *Let  $C$  have genus  $g \geq 2$ , and let  $W$  be a nontrivial stable bundle of degree zero and rank  $r$ . Set*

$$c = \min_{p \in C} h^0(C, W(p)).$$

If  $c \geq 1$ , then

$$r \geq gc + 2. \quad (6.1)$$

*Proof.* Set  $F = K_C \otimes W^\vee$ . Then  $F$  is stable of rank  $r$ , degree  $2r(g - 1)$ , and

$$h^0(F) = r(g - 1), \quad h^1(F) = h^0(W) = 0. \quad (6.2)$$

Let  $U \subset F$  be the saturation of the image of the evaluation map

$$H^0(F) \otimes \mathcal{O}_C \longrightarrow F,$$

and put  $s = \text{rk } U$ ,  $D = \deg U$ . At a point where  $h^0(W(p)) = c$ , the cokernel of evaluation in the fibre of  $F$  has dimension  $c$ : this follows from the exact sequence for  $F(-p)$ , because

$$H^1(F(-p))^\vee \simeq H^0(W(p)).$$

Consequently,

$$s = r - c, \quad h^0(U) = h^0(F) = r(g - 1). \quad (6.3)$$

Indeed, every global section of  $F$  lands generically in the evaluation image and therefore factors through its saturation  $U$ ; the reverse inclusion of section spaces is tautological.

The bundle  $U$  is generically globally generated. Its minimal Harder–Narasimhan quotient is therefore generically globally generated and has nonnegative degree; hence every Harder–Narasimhan slope of  $U$  is nonnegative. Its maximal-slope subbundle is a proper subbundle of the stable bundle  $F$ , so every such slope is strictly less than  $2g - 2$ . The vector-bundle Clifford inequality [5, Theorem 2.1], applied to the semistable Harder–Narasimhan factors and then added, gives

$$h^0(U) \leq s + \frac{D}{2}.$$

For a slope-zero factor, the same bound follows directly: generic global generation and degree zero make it trivial, so its number of sections is its rank. Thus no endpoint outside the cited positive-slope range is being used. Together with (6.3), this yields

$$D \geq 2r(g - 2) + 2c. \quad (6.4)$$

Stability of  $F$ , integrality of  $D$ , and  $s < r$  give

$$D \leq 2s(g - 1) - 1. \quad (6.5)$$

Equations (6.4) and (6.5) first imply  $r \geq gc + 1$ .

It remains to exclude the endpoint  $r = gc + 1$ . Then

$$s = (g - 1)c + 1,$$

and the two bounds for  $D$  differ by one. Thus

$$D = 2s(g - 1) - e, \quad e \in \{1, 2\}. \quad (6.6)$$

Define

$$E = K_C \otimes U^\vee.$$

Riemann–Roch, Serre duality, (6.3), and (6.6) give

$$\mathrm{rk} E = s, \quad \deg E = e, \quad h^0(E) = s + e - 1. \quad (6.7)$$

Writing  $Q = F/U$ , saturation makes  $Q$  locally free. Dualizing and tensoring by  $K_C$  gives a surjection

$$W \twoheadrightarrow E. \quad (6.8)$$

Since  $\mathrm{rk} E = s = r - c < r$ , every nonzero vector-bundle quotient of  $E$ , composed with (6.8), is a proper vector-bundle quotient of the stable degree-zero bundle  $W$ , and consequently has strictly positive degree.

I next show that  $E$  is generically globally generated. Let  $A \subset E$  be the saturation of its evaluation image, and suppose  $a = \mathrm{rk} A < s$ . The quotient  $E/A$  has positive degree, so

$$0 \leq \deg A \leq e - 1.$$

All sections of  $E$  factor through  $A$ , hence  $h^0(A) = h^0(E)$ . Choosing  $a$  generically independent sections of the generically generated bundle  $A$  gives

$$0 \longrightarrow \mathcal{O}_C^a \longrightarrow A \longrightarrow T_A \longrightarrow 0,$$

where  $T_A$  has length  $\deg A$ . It follows that

$$h^0(E) = h^0(A) \leq a + \deg A \leq s + e - 2,$$

contrary to (6.7). Thus  $E$  is generically globally generated.

Choose  $s$  generically independent global sections. Their determinant gives

$$0 \longrightarrow \mathcal{O}_C^s \longrightarrow E \longrightarrow T \longrightarrow 0, \quad \text{length } T = e. \quad (6.9)$$

Dualizing gives

$$0 \longrightarrow E^\vee \longrightarrow \mathcal{O}_C^s \longrightarrow \mathcal{E}xt^1(T, \mathcal{O}_C) \longrightarrow 0. \quad (6.10)$$

If  $s > e$ , the map  $\mathbf{C}^s \rightarrow H^0(\mathcal{E}xt^1(T, \mathcal{O}_C))$ , whose target has dimension  $e$ , has a nonzero kernel. The resulting section of  $E^\vee$  maps to a nonzero constant vector in  $\mathcal{O}_C^s$ , so it vanishes nowhere. Dualizing this line subbundle gives a quotient  $E \twoheadrightarrow \mathcal{O}_C$ , contradicting the positivity of all nonzero quotients of  $E$ . Therefore  $s \leq e$ .

Since  $s = (g - 1)c + 1 \geq 2$  and  $e \leq 2$ , the only remaining case is

$$s = e = 2, \quad g = 2, \quad c = 1. \quad (6.11)$$

Now  $E$  has rank two, degree two, and three global sections. The wedge map

$$\wedge^2 H^0(E) \longrightarrow H^0(\det E) \quad (6.12)$$

is injective. Indeed, every nonzero bivector in  $\wedge^2 \mathbf{C}^3$  is decomposable; a nonzero element of the kernel would give two independent sections contained in a line subbundle  $L \subset E$ . The quotient  $E/L$  has positive degree, so  $\deg L \leq 1$ , whereas  $h^0(L) \geq 2$ , impossible on a genus-two curve. Injectivity in (6.12) would imply  $h^0(\det E) \geq 3$ . A degree-two line bundle on a genus-two curve has at most two sections. This final contradiction excludes (6.11) and proves (6.1).  $\square$

**Corollary 6.2** (Degree and section bounds for covers). *Let  $f: X \rightarrow C$  be a connected finite étale cover of a genus- $g$  curve. If*

$$\min_{p \in C} h^0(X, \mathcal{O}_X(f^{-1}(p))) \geq k + 1 \quad (k \geq 1), \quad (6.13)$$

then

$$\deg f \geq gk + 3. \quad (6.14)$$

In particular, every connected Prill-exceptional étale cover has degree at least  $g + 3$ .

*Proof.* Decompose the unitary permutation bundle as

$$f_*\mathcal{O}_X \simeq \mathcal{O}_C \oplus \bigoplus_j W_j^{\oplus m_j}, \quad (6.15)$$

where the  $W_j$  are nontrivial stable degree-zero bundles. Put

$$c_j = \min_p h^0(W_j(p)).$$

The minimum of each upper-semicontinuous function is attained on a nonempty open subset. On the intersection of these finitely many open sets, (6.13) gives

$$\sum_j m_j c_j \geq k.$$

For every  $j$  with  $c_j > 0$ , theorem 6.1 gives  $\mathrm{rk} W_j \geq gc_j + 2$ . Hence

$$\deg f = 1 + \sum_j m_j \mathrm{rk} W_j \geq 1 + gk + 2 \sum_{c_j > 0} m_j \geq gk + 3.$$

□

**Corollary 6.3** (Galois lower bound). *If a connected Galois étale cover  $f: X \rightarrow C$  is Prill-exceptional, then its deck group  $G$  satisfies*

$$|G| \geq (g + 2)(g + 3). \quad (6.16)$$

*Proof.* Write the regular decomposition as

$$f_*\mathcal{O}_X \simeq \mathcal{O}_C \oplus \bigoplus_{\rho \neq \mathbf{1}} W_\rho^{\oplus r_\rho}, \quad r_\rho = \mathrm{rk} W_\rho.$$

On the common nonempty open set where every  $h^0(W_\rho(p))$  attains its minimum  $c_\rho$ , Prill-exceptionality gives  $\sum_{\rho \neq \mathbf{1}} r_\rho c_\rho \geq 1$ . Hence some nontrivial summand  $W = W_\rho$  has  $c = c_\rho > 0$ . If  $r = \mathrm{rk} W$ , then  $r \geq g + 2$  by theorem 6.1. The regular representation contains the trivial representation and  $W$  with multiplicity  $r$ , so  $|G| > r^2$ . The degree of a complex irreducible character divides the group order by the Frobenius divisibility theorem [16, §6.5]; hence  $|G|$  is a multiple of  $r$ , and

$$|G| \geq r(r + 1) \geq (g + 2)(g + 3).$$

□

**Corollary 6.4** (A numerical Putman–Wieland range). *A connected finite unramified cover of a closed genus- $g$  surface with degree at most  $g + 2$  cannot furnish a counterexample to the Putman–Wieland conjecture [13]. If the cover is Galois, the same conclusion holds whenever the deck group has order less than  $(g + 2)(g + 3)$ .*

*Proof.* Landesman and Litt prove that a closed Putman–Wieland counterexample is Prill-exceptional [6, Lemma 5.5]. Apply corollaries 6.2 and 6.3. □

*Remark 6.5* (No converse). Prill-exceptionality is only a necessary condition for a Putman–Wieland counterexample. The Heisenberg covers constructed here therefore do not by themselves produce finite homology orbits. The large-monodromy theorem of Landesman–Litt–Sawin [8] applies in a complementary range in which the base genus is large compared with the representation rank; the ranks  $n^g$  considered here lie far outside that range.

**6.1. The degree-five minimum in genus two.** The numerical lower bound leaves degree five as the first possible value in genus two. I first isolate its monodromy and theta obstruction, and then construct a symmetric example for which the obstruction vanishes.

**Lemma 6.6** (Rank-four no-jump lemma). *Let  $C$  have genus two, and let  $W$  be a nontrivial stable degree-zero bundle of rank four. If*

$$\min_{p \in C} h^0(C, W(p)) \geq 1,$$

*then*

$$h^0(C, W(p)) = 1 \quad \text{for every } p \in C. \quad (6.17)$$

*Proof.* The strict rank–corank inequality gives  $4 \geq 2c + 2$ , so the minimum is  $c = 1$ . Put

$$F = K_C \otimes W^\vee, \quad V = H^0(C, F).$$

Then  $\dim V = 4$ , and the evaluation map  $V \otimes \mathcal{O}_C \rightarrow F$  has generic rank three. Its kernel  $M$  is a line subbundle; set  $A = M^\vee$ . Let  $V_0 \subset V$  be the smallest constant subspace such that  $M \subset V_0 \otimes \mathcal{O}_C$ , and put  $t = \dim V_0$ . Minimality gives an injection  $V_0^\vee \hookrightarrow H^0(C, A)$ , and these sections generate  $A$ . If  $I_0$  is the image of  $V_0 \otimes \mathcal{O}_C \rightarrow F$  and  $U_0$  its saturation, then

$$\deg A = \deg I_0 \leq \deg U_0 \leq 2(t - 1) - 1, \quad (6.18)$$

because  $F$  is stable of slope two. The case  $t = 1$  would give a constant linear relation among the elements of  $V = H^0(F)$ , so  $t \geq 2$ .

On a genus-two curve, a globally generated nontrivial line bundle with at least  $t$  sections has degree at least 2, 4, 5 for  $t = 2, 3, 4$ , respectively. This contradicts (6.18) for  $t = 2, 3$ , so  $t = 4$  and  $\deg A \geq 5$ . Let  $I$  be the full evaluation image and  $U$  its saturation. Then  $\operatorname{rk} U = 3$ , stability gives  $\deg U \leq 5$ , and

$$\deg U = \deg I + \operatorname{length}(U/I) = \deg A + \operatorname{length}(U/I) \geq 5.$$

Consequently  $\deg A = \deg U = 5$  and  $U = I$ . Evaluation is therefore surjective onto  $U$  at every point, and its cokernel in  $F_p$  has dimension one for every  $p$ . Serre duality identifies this cokernel with  $H^0(C, W(p))^\vee$ , proving (6.17).  $\square$

**Proposition 6.7** (Degree-five monodromy reduction). *Let  $f: X \rightarrow C$  be a connected degree-five étale cover of a genus-two curve. If every fibre moves in a pencil, then the monodromy group is  $A_5$  in its natural five-point action. The augmentation bundle  $W$  is a stable oriented orthogonal bundle of rank four and satisfies*

$$h^0(C, W(p)) = 1 \quad (p \in C), \quad w_1(W) = 0, \quad \langle w_2(W), [C] \rangle = 1. \quad (6.19)$$

*Proof.* Decompose the augmentation local system into irreducible unitary summands. On the common open set where all section dimensions are minimal, Prill-exceptionality forces one summand  $E$  to satisfy  $\min_p h^0(E(p)) > 0$ . By theorem 6.1,  $\operatorname{rk} E \geq 4$ . Since the full augmentation rank is four, the augmentation representation is irreducible. It is therefore stable, and lemma 6.6 gives the first assertion in (6.19).

An irreducible augmentation representation is equivalent to two-transitivity of the five-point action: the number of orbits on ordered pairs is  $\dim \text{End}_G(\mathbf{C}^5)$ . The two-transitive subgroups of  $S_5$  are

$$\text{AGL}_1(\mathbf{F}_5), \quad A_5, \quad S_5. \quad (6.20)$$

For completeness, two-transitivity makes  $|G|$  divisible by  $5 \cdot 4$ . If  $|G| = 20$ , the action is sharply two-transitive and  $G = C_5 \rtimes C_4$ . Subgroups of orders 40 and 60 not equal to  $A_5$  are excluded by intersecting with the simple group  $A_5$ , and order 120 gives  $S_5$ .

It remains to use the orthogonal structure. For a real orthogonal bundle  $E_{\mathbf{R}}$  of rank  $r$  and a theta characteristic  $\kappa$ , the mod-two index formula [2] is

$$h^0(C, E_{\mathbf{R}} \otimes_{\mathbf{R}} \mathbf{C} \otimes \kappa) \equiv rh^0(C, \kappa) + q_{\kappa}(w_1(E_{\mathbf{R}})) + \langle w_2(E_{\mathbf{R}}), [C] \rangle \pmod{2}, \quad (6.21)$$

where  $q_{\kappa}$  is the Riemann–Mumford quadratic refinement. One may derive (6.21) from the mod-two index theorem by the splitting principle: for real line classes  $a_i$ , use  $q_{\kappa}(\sum a_i) = \sum q_{\kappa}(a_i) + \sum_{i < j} a_i \smile a_j$ .

The six Weierstrass points give the six odd theta characteristics  $\kappa_i = \mathcal{O}_C(w_i)$ . Equation (6.17) and (6.21) imply

$$q_{\kappa_i}(w_1(W)) + \langle w_2(W), [C] \rangle = 1 \quad (1 \leq i \leq 6). \quad (6.22)$$

For every nonzero  $a \in H^1(C, \mathbf{F}_2)$ , the values  $q_{\kappa}(a)$ , as  $q_{\kappa}$  ranges over the six odd quadratic forms, are one four times and zero twice. This follows by the transitivity of  $\text{Sp}_4(\mathbf{F}_2)$  on nonzero vectors and a direct count for one symplectic basis vector. Hence the left side of (6.22) can be constant only if  $w_1(W) = 0$ , and then  $w_2(W) = 1$ .

Finally, the determinant of the augmentation representation is the sign of the five-point permutation action. Thus  $w_1(W) = 0$  forces  $G \subset A_5$ , and (6.20) gives  $G = A_5$ .  $\square$

**Proposition 6.8** (The remaining theta obstruction). *Let  $\rho: \pi_1(C) \twoheadrightarrow A_5$  be a genus-two surface representation whose natural oriented augmentation bundle  $W_{\rho}$  has  $w_2 = 1$ . Let  $\Theta \simeq C \subset J^1(C)$  be the Abel theta divisor, and let  $s_{\rho}$  be the determinant-of-cohomology theta section of  $W_{\rho}$ , allowing  $s_{\rho} = 0$ . After choosing the standard theta-line identification up to scalar and a nonzero Wronskian normalization, there is a unique one-form  $\omega_{\rho} \in H^0(C, K_C)$  such that*

$$s_{\rho}|_{\Theta} = \text{Wr}_C \omega_{\rho}, \quad (6.23)$$

where a nonzero Wronskian  $\text{Wr}_C \in H^0(C, K_C^3)$  has been chosen; it is well-defined up to scalar and has zero divisor equal to the six Weierstrass points. The form  $\omega_{\rho}$  is unique relative to this normalization, and its vanishing is intrinsic. The associated degree-five cover is Prill-exceptional if and only if  $\omega_{\rho} = 0$ .

Moreover, it suffices to verify the Prill inequality at three distinct non-Weierstrass points: three such zeros force  $\omega_{\rho} = 0$ , since a nonzero holomorphic one-form on a genus-two curve has a zero divisor of degree two.

*Proof.* For an oriented orthogonal bundle of rank four, Beauville’s parity formula and theta-map lemma [4, (1.3), Lemma 1.4] place the theta section in the odd eigenspace  $H^0(J^1(C), \mathcal{O}(4\Theta))^-$  when  $w_2 = 1$ . In particular,  $h^0(C, W_{\rho}(w_i))$  is odd at every Weierstrass point, so  $s_{\rho}|_{\Theta}$  vanishes at all six points. Adjunction gives  $\mathcal{O}_{J^1(C)}(\Theta)|_{\Theta} \simeq K_C$ , hence  $s_{\rho}|_{\Theta} \in H^0(C, K_C^4)$ . After choosing a nonzero Wronskian normalization, division gives the corresponding unique  $\omega_{\rho} \in H^0(C, K_C)$  in (6.23).

By the defining property of the determinant-of-cohomology section,

$$s_{\rho}(\mathcal{O}_C(p)) = 0 \iff H^0(C, W_{\rho}(p)) \neq 0.$$

Thus every fibre moves exactly when the restriction in (6.23) is identically zero, which is equivalent to  $\omega_{\rho} = 0$ . The final assertion follows from  $\deg K_C = 2$ .  $\square$

**Lemma 6.9** (Equivariant form of the obstruction). *Let a finite group  $H$  act on a genus-two curve  $C$ , and suppose that the oriented orthogonal bundle  $W_\rho$  in [proposition 6.8](#) is  $H$ -equivariant. Put*

$$V_C = H^0(C, K_C), \quad D_\rho = \det R\Gamma(C, W_\rho),$$

*and let  $\Delta_\rho$  be the one-dimensional  $H$ -character carried by the  $H$ -linearized trivial determinant line  $\det W_\rho \simeq \mathcal{O}_C$ . Define the character line*

$$\Xi_\rho = D_\rho^{-1} \otimes \Delta_\rho^{-1} \otimes \det V_C. \quad (6.24)$$

*Then the intrinsic obstruction is an invariant tensor*

$$\Omega_\rho \in (V_C \otimes \Xi_\rho)^H. \quad (6.25)$$

*In particular, if the invariant space in (6.25) is zero, then the associated degree-five cover is Prill-exceptional.*

*Proof.* Let  $a: C \rightarrow J^1(C)$  be  $p \mapsto \mathcal{O}_C(p)$ , and use the inverse determinant-of-cohomology convention for the theta line  $\lambda_\rho$ . The exact sequence

$$0 \longrightarrow W_\rho \longrightarrow W_\rho(p) \longrightarrow W_{\rho,p} \otimes T_p C \longrightarrow 0$$

gives the  $H$ -equivariant identity

$$a^* \lambda_\rho \simeq D_\rho^{-1} \otimes \Delta_\rho^{-1} \otimes K_C^4. \quad (6.26)$$

The basis-free canonical Wronskian is a tensor

$$\mathfrak{W}_C \in H^0(C, K_C^3) \otimes (\det V_C)^{-1}.$$

Dividing the invariant restriction  $a^* s_\rho$  by  $\mathfrak{W}_C$  therefore gives precisely an invariant element of  $V_C \otimes \Xi_\rho$ . After bases are chosen in the character lines, this is the one-form  $\omega_\rho$  of [proposition 6.8](#); only its vanishing is independent of those choices.  $\square$

**Theorem 6.10** (A degree-five Prill cover and the genus-two minimum). *Let*

$$C: y^2 = x^5 - 1.$$

*There exists a connected degree-five étale cover  $f: X \rightarrow C$  such that*

$$h^0(X, \mathcal{O}_X(f^{-1}(p))) \geq 2 \quad \text{for every } p \in C. \quad (6.27)$$

*Its monodromy group is  $A_5$  in the natural five-point action,  $g(X) = 6$ , and its  $A_5$ -Galois closure has degree 60 and genus 61. Consequently*

$$\min \left\{ \deg f' \mid \begin{array}{l} C' \text{ is a genus-two curve,} \\ f': X' \rightarrow C' \text{ is connected and Prill-exceptional} \end{array} \right\} = \boxed{5}. \quad (6.28)$$

*Proof.* Fix a primitive fifth root  $\zeta$ , and let  $\sigma(x, y) = (\zeta x, y)$ . The fixed points are  $P_\pm = (0, \pm i)$  and the point  $P_\infty$  at infinity. The tangent multipliers of  $\sigma$  are  $\zeta, \zeta, \zeta^2$ : at  $P_\pm$  use  $x$ , while at infinity one may use  $t = x^2/y$ , for which  $\sigma(t) = \zeta^2 t$ . Hence  $C/\langle \sigma \rangle \simeq \mathbf{P}^1$  has signature  $(0; 5, 5, 5)$ , with orbifold epimorphism

$$\Delta(5, 5, 5) = \langle x_1, x_2, x_3 \mid x_i^5 = x_1 x_2 x_3 = 1 \rangle \longrightarrow \langle \sigma \rangle, \quad (x_1, x_2, x_3) \longmapsto (\sigma, \sigma, \sigma^3). \quad (6.29)$$

In  $SL_2(\mathbf{F}_5)$ , take

$$U_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad U_2 = \begin{pmatrix} 3 & 1 \\ 1 & 4 \end{pmatrix}, \quad U_3 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}. \quad (6.30)$$

Direct multiplication gives

$$U_1^5 = U_2^5 = U_3^5 = I, \quad U_1 U_2 U_3 = -I. \quad (6.31)$$

Moreover  $U_1, U_3$  generate  $SL_2(\mathbf{F}_5)$ , since they are the elementary upper and lower unipotent matrices. If  $\bar{U}_i$  denotes the image in  $PSL_2(\mathbf{F}_5) \simeq A_5$ , then

$$x_1 \mapsto (\bar{U}_1, \sigma), \quad x_2 \mapsto (\bar{U}_2, \sigma), \quad x_3 \mapsto (\bar{U}_3, \sigma^3) \quad (6.32)$$

defines a homomorphism onto  $A_5 \times C_5$ . Indeed, both projections are surjective, and  $A_5$  and  $C_5$  have no common nontrivial quotient. Its restriction to the index-five surface kernel in (6.29) is therefore surjective onto  $A_5$ . The natural five-point action gives a connected degree-five étale cover and an equivariant augmentation bundle  $W_\rho$ .

I next compute its spin obstruction. The pullback of  $\text{Spin}(4) \rightarrow \text{SO}(4)$  under the natural augmentation embedding is the Schur cover

$$2 \cdot A_5 \simeq SL_2(\mathbf{F}_5).$$

It is non-split: a splitting would embed the simple group  $A_5$  in  $\text{Spin}(4) = SU(2) \times SU(2)$ , so one projection would embed  $A_5$  in  $SU(2)$ , which is impossible because  $SU(2)$  has only one element of order two whereas  $A_5$  has fifteen. Each order-five element of  $PSL_2(\mathbf{F}_5)$  has a unique lift whose fifth power is  $I$ ; the other lift has fifth power  $-I$ . Thus the orbifold relations force the three lifts in (6.30), whose product is  $-I$  by (6.31). The orbifold representation has a nonzero spin-lifting class. Restriction to the surface subgroup preserves nonvanishing: with  $\mathbf{F}_2$ -coefficients,  $\text{cor} \circ \text{res}$  is multiplication by the index 5, hence is the identity. Consequently

$$\langle w_2(W_\rho), [C] \rangle = 1. \quad (6.33)$$

It remains to kill the holomorphic obstruction. Write  $\chi(\sigma) = \zeta$ . Pullback acts on canonical forms by

$$V_C = H^0(C, K_C) = \left\langle \frac{dx}{y}, \frac{x dx}{y} \right\rangle \simeq \chi \oplus \chi^2, \quad \det V_C = \chi^3. \quad (6.34)$$

The augmentation character has value  $-1$  on every nontrivial five-cycle. At infinity the action of  $\sigma$  on the fibre is the square of the third inertia element, which is still a nontrivial five-cycle. Since  $H^0(C, W_\rho) = 0$ , Serre duality gives  $H^1(C, K_C \otimes W_\rho) = 0$ . The holomorphic Lefschetz formula [3], with pullback linearization and denominator  $1 - \lambda$ , gives for  $1 \leq k \leq 4$

$$\text{Tr}(\sigma^k \mid H^0(C, K_C \otimes W_\rho)) = - \left( \frac{2\zeta^k}{1 - \zeta^k} + \frac{\zeta^{2k}}{1 - \zeta^{2k}} \right). \quad (6.35)$$

For a nontrivial fifth root  $q$ ,

$$\frac{2q}{1 - q} + \frac{q^2}{1 - q^2} = q + q^2 - 1.$$

It follows that

$$H^0(C, K_C \otimes W_\rho) \simeq \mathbf{1}^{\oplus 2} \oplus \chi^3 \oplus \chi^4. \quad (6.36)$$

With the convention  $D_\rho = \det H^0(C, W_\rho) \otimes \det H^1(C, W_\rho)^{-1}$ , Serre duality and (6.36) give

$$D_\rho = \chi^2.$$

The representation  $A_5 \times C_5 \rightarrow \text{SO}(4)$  factors through  $A_5$ , so its determinant character is trivial and  $\Delta_\rho = \mathbf{1}$ . Therefore

$$\Xi_\rho = D_\rho^{-1} \otimes \Delta_\rho^{-1} \otimes \det V_C = \chi, \quad V_C \otimes \Xi_\rho = \chi^2 \oplus \chi^3. \quad (6.37)$$

The latter representation has no invariants. By lemma 6.9,  $\Omega_\rho = 0$ , and then proposition 6.8 gives  $H^0(C, W_\rho(p)) \neq 0$  for every  $p \in C$ . Since

$$f_* \mathcal{O}_X \simeq \mathcal{O}_C \oplus W_\rho,$$

this is exactly (6.27).

The genus statements follow from the unramified Riemann–Hurwitz formula. Finally, [corollary 6.2](#) gives degree at least  $2 + 3 = 5$  for every connected genus-two Prill-exceptional cover, so the constructed degree-five cover proves [\(6.28\)](#).  $\square$

*Remark 6.11* (Reproducible finite certificates). The supplementary GAP certificate and independent Python character check verify the finite group generation, the lifted surface relation, and the  $C_5$ -character decomposition used above. The construction only forces the generalized theta divisor to contain the Abel curve  $\Theta \simeq C$ ; it does not assert vanishing over all of  $\text{Pic}^1(C)$ .

## 7. CONSEQUENCES FOR GENERIC GENERATION AND HORIZONTAL VANISHING

I record the exact implications for the questions formulated in [\[9, §6.3\]](#). Throughout this section  $g \geq 3$ , and I take  $n = g$  in [theorem 3.3](#). Write  $V$  for the resulting rank- $g^g$  flat bundle.

**Proposition 7.1** (Failure of generic global generation). *The fixed irreducible finite-unitary representation dual to  $\rho_{g,g}$  has the following property for every marked complex structure:*

$$K_C \otimes V^\vee \text{ is not generated by global sections at any point of } C. \quad (7.1)$$

Consequently, [Conjectures 6.3.5 and 6.3.11](#) of [\[9\]](#) are false.

*Proof.* Since  $H^0(V) = 0$ , Serre duality gives  $H^1(K_C \otimes V^\vee) = 0$ . From

$$0 \longrightarrow K_C \otimes V^\vee(-p) \longrightarrow K_C \otimes V^\vee \longrightarrow (K_C \otimes V^\vee)|_p \longrightarrow 0,$$

the dual of the evaluation cokernel at  $p$  is

$$H^1(K_C \otimes V^\vee(-p))^\vee \simeq H^0(V(p)). \quad (7.2)$$

This space is nonzero for every  $p$  by [theorem 3.3](#). Hence evaluation fails to be surjective at every point, for every complex structure. The cited conjectures predict generic, respectively global, generation for every fixed unitary representation on a generic complex structure, so this fixed representation refutes both predictions.  $\square$

**Proposition 7.2** (Failure of horizontal generic vanishing). *In the unpunctured case  $D = \emptyset$ , the fixed monodromy  $\rho_{g,g}$  satisfies*

$$H^0(C, V \otimes L) \neq 0 \text{ for every marked complex structure and every } L \in \text{Pic}^1(C). \quad (7.3)$$

Thus both the weak and strong forms of [Conjecture 6.3.17](#) in [\[9\]](#) fail.

*Proof.* Equation [\(7.3\)](#) is [theorem 3.3](#) with  $n = g$ . Apply [Conjecture 6.3.17](#) with  $D = \emptyset$ . Its nondecreasing function has value 1 at  $g = 3$ , hence value at least 1 for every  $g \geq 3$ . Taking the degree-one divisor  $Z = p$ , the nonvanishing  $H^0(C, V(p)) \neq 0$  for every  $p$ , and for every isomonodromic deformation, contradicts both the weak and strong forms.  $\square$

**Corollary 7.3** (Status of the adjacent questions). *The construction has the following consequences.*

- (i) *It answers Prill’s problem affirmatively for every base genus  $g \geq 2$ , with every fibre moving.*
- (ii) *It disproves the unpunctured specialization, [Conjecture 6.3.5](#), of the parabolic generic-generation [Conjecture 6.3.4](#).*

*Proof.* The first statement is [corollary 2.5](#). The second is [proposition 7.1](#) together with the logical relation among the conjectures stated in [\[9, §6.3\]](#).  $\square$

*Remark 7.4* (Scope of the consequences). The construction is unpunctured and supplies no conclusion for the genuinely punctured cases of Conjecture 6.3.4. No implication is established here from these support-locus statements to a Putman–Wieland or Ivanov counterexample, a constant Prym variation, or an isotrivial Jacobian factor; see [remark 6.5](#).

## 8. ABELIAN SUPPORT BLOCKS AND LAGRANGIAN THETA DESCENT

This section isolates the construction from the special choice  $A = J(C)$ . It produces abelian blocks in support loci whenever a curve maps homologically onto an abelian variety.

Let  $A$  be a complex abelian variety of dimension  $b$ , let  $\hat{A} = \text{Pic}^0(A)$ , and let  $N$  be an ample line bundle on  $A$ . For  $a \in A = \text{Pic}^0(\hat{A})$ , write  $P_a = \mathcal{P}|_{\hat{A} \times \{a\}}$  for the corresponding topologically trivial line bundle on  $\hat{A}$ . Write

$$\phi_N: A \longrightarrow \hat{A}, \quad r = h^0(A, N), \quad K(N) = \ker \phi_N. \quad (8.1)$$

Thus  $|K(N)| = r^2$ . Let  $N_0$  be the exponent of  $K(N)$ , and define the Fourier–Mukai bundle

$$F_N = \Phi_{\mathcal{P}}(N^{-1})[b] \quad \text{on } \hat{A}. \quad (8.2)$$

For a morphism  $u: C \rightarrow \hat{A}$ , put  $R_{u,N} = u^*F_N$ .

**Theorem 8.1** (Lagrangian theta descent and abelian support). *Assume that*

$$u_*: H_1(C, \mathbf{Z}) \rightarrow H_1(\hat{A}, \mathbf{Z}) \quad (8.3)$$

*is surjective. Then the following statements hold.*

- (i) *The bundle  $R_{u,N}$  is stable of rank  $r$ . Its projective monodromy is the irreducible theta representation of the finite symplectic module  $K(N)$ , with the central weight fixed by the dual section factor in (8.10).*
- (ii) *Let  $\beta_N \in H^2(\hat{A}, \mathbf{Q})$  be defined by*

$$\phi_N^* \beta_N = c_1(N). \quad (8.4)$$

*Then*

$$\mu(R_{u,N}) = \int_C u^* \beta_N. \quad (8.5)$$

- (iii) *If*

$$B_u = \text{im} \left( u^*: A = \text{Pic}^0(\hat{A}) \longrightarrow \text{Pic}^0(C) \right), \quad (8.6)$$

*then*

$$B_u \subset V_0^0(R_{u,N}). \quad (8.7)$$

*Here  $V_0^0(R_{u,N}) = \{\alpha \in \text{Pic}^0(C) : H^0(C, R_{u,N} \otimes \alpha) \neq 0\}$ .*

- (iv) *After choosing a linearized maximal isotropic subgroup of  $K(N)$ , there is a connected degree- $r$  étale cover  $q: C_L \rightarrow C$  and a line bundle  $\mathcal{L}$  on  $C_L$  such that*

$$R_{u,N} \simeq q_* \mathcal{L}, \quad H^i(C, R_{u,N} \otimes A_C) \simeq H^i(C_L, \mathcal{L} \otimes q^* A_C) \quad (8.8)$$

*for every line bundle  $A_C$  on  $C$  and  $i = 0, 1$ . Moreover,*

$$\deg \mathcal{L} = \deg R_{u,N} = r \mu(R_{u,N}). \quad (8.9)$$

*Proof.* Mukai’s isogeny formula, with its theta-group equivariance, gives

$$\phi_N^* F_N \simeq (-1_A)^* N \otimes H^0(A, N)^\vee. \quad (8.10)$$

The fibre product

$$\tilde{C} = C \times_{\hat{A}, \phi_N} A$$

is connected by (8.3). Pulling (8.10) to  $\tilde{C}$  shows that  $R_{u,N}$  has the stated projective monodromy.

The same identity makes the pullback of  $R_{u,N}$  a direct sum of copies of one line bundle, so  $R_{u,N}$  is semistable. To prove stability, suppose a proper subbundle has the same slope. After pullback to  $\tilde{C}$  and tensoring by the inverse of the line factor in (8.10), it becomes a degree-zero subbundle  $G$  of  $\mathcal{O}_{\tilde{C}} \otimes H^0(A, N)^\vee$ . Such a subbundle is constant. Indeed,  $G^\vee$  is a globally generated degree-zero quotient of a trivial bundle. Choosing  $\text{rk } G$  generically independent sections produces a torsion quotient of length  $\deg G^\vee = 0$ , so  $G^\vee \simeq \mathcal{O}_{\tilde{C}}^{\oplus \text{rk } G}$ , and the inclusion of  $G$  is induced by a constant subspace  $W \subset H^0(A, N)^\vee$ . Equivariance of the deck descent forces  $W$  to be invariant under the full projective Schrödinger action, contrary to irreducibility unless  $W = 0$  or  $W = H^0(A, N)^\vee$ . This proves (i).

Taking first Chern classes in (8.10) gives

$$\phi_N^* \left( \frac{c_1(F_N)}{r} \right) = c_1(N).$$

Pullback to  $C$  proves (ii).

For (iii), Mukai inversion [10] gives

$$\hat{\Phi}_{\mathcal{P}}(F_N) \simeq (-1_A)^* N^{-1}$$

in degree zero. Thus  $F_N$  satisfies  $\text{IT}_0$ , has Euler characteristic one, and base change gives

$$h^0(\hat{A}, F_N \otimes P_a) = 1, \quad H^i(\hat{A}, F_N \otimes P_a) = 0 \quad (i > 0)$$

for every  $a \in A$ . These one-dimensional spaces are the fibres of the transform line. After trivializing that line over a nonempty open subset of  $A$ , base change produces an evaluation morphism. It is nonzero somewhere, since each fibre is a nonzero global section; hence it is nonzero at some point  $x_0 \in \hat{A}$ . Since  $F_N$  is simple semihomogeneous, translation by every  $t \in \hat{A}$  has the form

$$t^* F_N \simeq F_N \otimes P_{\psi(t)} \tag{8.11}$$

for a suitable  $\psi(t) \in A$ . Translating the preceding family shows that, for every prescribed  $x \in \hat{A}$ , there is a nonempty open set  $U_x \subset A$  on which the corresponding section is nonzero at  $x$ .

Fix  $p \in C$  and take  $x = u(p)$ . For  $a \in U_x$ , restriction gives

$$H^0(C, R_{u,N} \otimes u^* P_a) \neq 0.$$

The locus of  $a$  satisfying this inequality is closed by upper semicontinuity. It contains a nonempty open subset of the irreducible variety  $A$ , and hence it is all of  $A$ . This proves (8.7). This continuous-generation argument is the general ample-polarization form of Raynaud's construction [14, §3].

Finally, choose a maximal isotropic theta subgroup. Standard Lagrangian descent for theta groups, in the form used in [15, Proposition 2.1(2)], gives an abelian variety  $\hat{A}_L$ , an isogeny  $\tau: \hat{A}_L \rightarrow \hat{A}$  of degree  $r$ , and a line bundle  $N_L$  such that

$$F_N \simeq \tau_* N_L.$$

The base change  $C_L = C \times_{\hat{A}} \hat{A}_L$  is connected by (8.3). Flat base change proves (8.8). Since  $q$  is étale, Riemann–Roch gives  $\deg \mathcal{L} = \deg q_* \mathcal{L}$ , proving (8.9).  $\square$

**Theorem 8.2** (Integral-slope Heisenberg untwisting). *In the setting of theorem 8.1, suppose*

$$\mu = \mu(R_{u,N}) \in \mathbf{Z}. \tag{8.12}$$

Assume in addition that  $K(N) \neq 0$ , equivalently  $N_0 \geq 2$ . Then there is a finite irreducible unitary local system  $V$  of rank  $r$ , a line bundle  $M \in \text{Pic}^\mu(C)$ , and a connected étale cover

$$f: Z \longrightarrow C, \quad \deg f = N_0 r, \quad (8.13)$$

such that

$$R_{u,N} \simeq V \otimes M, \quad f_* \mathcal{O}_Z \simeq \mathcal{O}_C \oplus V \oplus E, \quad (8.14)$$

Writing  $M + B_u := \{M \otimes \alpha : \alpha \in B_u\} \subset \text{Pic}^\mu(C)$ , one has

$$h^0(Z, f^* L) \geq h^0(C, L) + 1 \quad \text{for every } L \in M + B_u. \quad (8.15)$$

The Galois closure has degree  $N_0 r^2$ . Equivalently,

$$f^*(M + B_u) \subset W_{N_0 r \mu}^0(Z). \quad (8.16)$$

*Proof.* For a projectively flat bundle  $E$  of rank  $r$  and degree  $d$ , the exact sequence

$$1 \longrightarrow \mu_r \longrightarrow SL_r \longrightarrow PGL_r \longrightarrow 1$$

associates to  $\mathbf{P}(E)$  an obstruction  $o(E) \in H^2(C, \mu_r)$ . Under  $H^2(C, \mu_r) \simeq \mathbf{Z}/r$ , this class is  $d \bmod r$ : twisting a  $GL_r$ -lift by a line bundle changes its determinant degree by a multiple of  $r$ , and determinant reduction gives the converse. Under the embedding  $\mu_r \subset \mathbf{C}^*$ , evaluation on  $[C]$  is therefore

$$\exp(\pm 2\pi i d/r), \quad (8.17)$$

where the sign depends only on the commutator convention. Here  $d/r = \mu$ , so (8.12) kills the surface obstruction. The projective theta class is determined by its commutator because  $H^2(K(N), \mathbf{C}^*) \simeq \text{Hom}(\wedge^2 K(N), \mathbf{C}^*)$ . It may therefore be represented by the split finite model  $H_{\text{sp}}(K(N))$  constructed in theorem 4.2:

$$1 \longrightarrow \mu_{N_0} \longrightarrow H_{\text{sp}}(K(N)) \longrightarrow K(N) \longrightarrow 1.$$

Choosing lifts of surface generators in this model, the vanishing obstruction makes their product of commutators one; hence lemma 2.7 gives a homomorphism  $\pi_1(C) \rightarrow H_{\text{sp}}(K(N))$ . Its Schrödinger representation, with the central weight matching (8.10), defines  $V$ .

The finite lift is surjective. Indeed, its projective image is all of  $K(N)$ , while commutators of lifts generate  $\mu_{N_0}$  because the theta pairing is perfect and has exponent  $N_0$ .

Choose  $M_0 \in \text{Pic}^\mu(C)$ . The bundles  $R_{u,N}$  and  $V \otimes M_0$  have the same projectivization and the same degree. By lemma 3.2,  $R_{u,N} \simeq V \otimes M_0 \otimes \alpha$  for some  $\alpha \in \text{Pic}^0(C)$ . Setting  $M = M_0 \otimes \alpha$  gives the first identity in (8.14).

Equation (8.7) now says

$$H^0(C, V \otimes L) \neq 0 \quad (L \in M + B_u). \quad (8.18)$$

Apply theorem 4.2 to this surjection. It produces the degree- $N_0 r$  cover, places  $\mathcal{O}_C \oplus V$  in its permutation local system, and gives the core-free Galois closure of degree  $N_0 r^2$ . The projection formula and (8.18) prove (8.15); its rank-one reformulation is (8.16).  $\square$

**Corollary 8.3** (Prym-direction Brill–Noether block). *Let  $P \subset J(Y)$  be an abelian subvariety of dimension  $b$ , endowed with a principal polarization  $\Xi$  not assumed to be the restriction of the theta polarization of  $J(Y)$ . Use the Jacobian principal polarization to identify  $J(Y) \simeq J(Y)^\vee$ , and let*

$$q_P: J(Y) \longrightarrow P^\vee$$

*be dual to the inclusion, and let  $u_P = q_P \circ \text{AJ}_Y$ . If  $\hat{\Xi}$  is the dual principal class, assume*

$$D = \deg(u_P^* \hat{\Xi}) \geq 2. \quad (8.19)$$

Then there is an  $M \in \text{Pic}^1(Y)$  and a connected étale cover

$$f: Z \longrightarrow Y, \quad \deg f = D^{b+1}, \quad (8.20)$$

such that

$$f^*(M + u_P^*P) \subset W_{D^{b+1}}^0(Z). \quad (8.21)$$

Its Heisenberg Galois closure has degree  $D^{2b+1}$ .

*Proof.* The kernel of  $q_P$  is connected, so  $u_P$  is surjective on integral first homology. Apply [theorem 8.1](#) with  $A = P$  and  $N = \Xi^{\otimes D}$ . Then

$$r = D^b, \quad K(N) = P[D], \quad N_0 = D.$$

Since

$$\beta_N = \frac{1}{D} c_1(\widehat{\Xi}),$$

[equations \(8.5\)](#) and [\(8.19\)](#) give  $\mu(R_{u_P, N}) = 1$ . Now apply [theorem 8.2](#).  $\square$

*Remark 8.4* (Prym boundary). If  $P$  is a proper abelian subvariety of  $J(Y)$ , the translate  $M + u_P^*P$  is only a proper abelian block in  $\text{Pic}^1(Y)$ . It need not contain the Abel curve  $p \mapsto \mathcal{O}_Y(p)$ . Thus [corollary 8.3](#) is a Brill–Noether statement, not a full Prill statement and not a claim about finite higher-Prym monodromy. The statement applies in particular to principally polarized Prym varieties; no principal polarization is asserted for an arbitrary ramified Prym.

## 9. EXACT BOUNDARIES OF THE CONSTRUCTION

I finish by isolating the hypotheses that cannot be removed formally.

**Proposition 9.1** (Integrality is the flat lifting condition). *Let  $E$  be a projectively flat bundle of rank  $r$  and degree  $d$  whose projective monodromy is a finite theta representation. Its projective representation has a flat linear lift on  $C$  if and only if*

$$r \mid d. \quad (9.1)$$

*If the scalar obstruction has order  $s$ , then pullback along a connected finite étale cover of degree  $\ell$  kills it if and only if  $s \mid \ell$ .*

*Proof.* The obstruction class is  $d \bmod r \in H^2(C, \mu_r)$ , as proved above, so it vanishes exactly when  $r \mid d$ . If  $p: C' \rightarrow C$  has degree  $\ell$ , then  $p_*[C'] = \ell[C]$ ; hence evaluation of  $p^*o(E)$  is the  $\ell$ -th power of the original scalar. A scalar of exact order  $s$  is killed precisely when  $s \mid \ell$ .  $\square$

This proposition concerns a flat lift of the projective representation; the bundle  $E$  itself is flat only when  $d = 0$ . A lift inside the finite split theta model is obtained, when needed, by the construction in [theorem 4.2](#).

For the Raynaud projective bundle at level  $n$ , the scalar is  $\zeta^{-g}$ , of order

$$\frac{n}{\gcd(n, g)}. \quad (9.2)$$

Thus  $n \mid g$  is not a convenient sufficient assumption; it is the exact condition for a flat scalar lift on the original surface. Killing the scalar after a base change does not transfer the full support property from  $\text{Pic}(C)$  to the full Picard variety of the new curve.

*Remark 9.2* (Marked versus characteristic). The homomorphism  $\varphi_{g,n}$  and every cover in the tower are fixed after choosing a marking. Their kernels are not claimed to be characteristic subgroups of  $\pi_g$ . The projective theta class is symplectic, but linear lifts can be permuted by scalar characters under the mapping class group.

*Remark 9.3* (Even and composite levels). No primeness or parity hypothesis on  $n$  appears in the construction. The trace calculation, isotropic order bound, and exact degree–multiplicity law are valid over  $\mathbf{Z}/n$ . Even-level quadratic refinements become relevant only when extending the symplectic group to a prescribed metaplectic action. The fixed marked projective class is already determined by its commutator pairing.

*Remark 9.4* (The three geometric outputs). The method separates three different kinds of information:

$$\begin{aligned} \text{theta-equivariant Fourier–Mukai splitting} &\implies \text{finite projective monodromy,} \\ \text{integral slope and scalar transport} &\implies \text{a fixed flat support block,} \\ \text{Lagrangian induction and cyclic untwisting} &\implies \text{rank-one geometry on a finite cover.} \end{aligned} \tag{9.3}$$

The last step is what lowers the ordinary cover degree from  $n^{2g+1}$  to  $n^{g+1}$ . The middle step is what makes the construction uniform over marked complex structures. Neither step alone implies fixed-part monodromy.

## REFERENCES

- [1] E. Arbarello, M. Cornalba, P. A. Griffiths, and J. Harris, *Geometry of algebraic curves. Vol. I*, Grundlehren Math. Wiss. **267**, Springer, New York, 1985.
- [2] M. F. Atiyah, Riemann surfaces and spin structures, *Ann. Sci. École Norm. Sup. (4)* **4** (1971), 47–62.
- [3] M. F. Atiyah and R. Bott, A Lefschetz fixed point formula for elliptic complexes. II. Applications, *Ann. of Math. (2)* **88** (1968), 451–491.
- [4] A. Beauville, Orthogonal bundles on curves and theta functions, *Ann. Inst. Fourier (Grenoble)* **56** (2006), 1405–1418.
- [5] L. Brambila-Paz, I. Grzegarczyk, and P. E. Newstead, Geography of Brill–Noether loci for small slopes, *J. Algebraic Geom.* **6** (1997), 645–669.
- [6] A. Landesman and D. Litt, Applications of the algebraic geometry of the Putman–Wieland conjecture, *Proc. Lond. Math. Soc. (3)* **127** (2023), 116–133.
- [7] A. Landesman and D. Litt, Prill’s problem, *Algebraic Geometry* **11** (2024), 290–295.
- [8] A. Landesman, D. Litt, and W. Sawin, Big monodromy for higher Prym representations, *Geom. Topol.* **29** (2025), 2733–2782.
- [9] D. Litt, Motives, mapping class groups, and monodromy, arXiv:2409.02234v1, 2024.
- [10] S. Mukai, Duality between  $D(X)$  and  $D(\hat{X})$  with its application to Picard sheaves, *Nagoya Math. J.* **81** (1981), 153–175.
- [11] D. Mumford, *Tata lectures on theta. I*, Progress in Mathematics **28**, Birkhäuser, Boston, 1983.
- [12] M. S. Narasimhan and C. S. Seshadri, Stable and unitary vector bundles on a compact Riemann surface, *Ann. of Math. (2)* **82** (1965), 540–567.
- [13] A. Putman and B. Wieland, Abelian quotients of subgroups of the mapping class group and higher Prym representations, *J. Lond. Math. Soc. (2)* **88** (2013), 79–96.
- [14] M. Raynaud, Sections des fibrés vectoriels sur une courbe, *Bull. Soc. Math. France* **110** (1982), 103–125.
- [15] O. Schneider, Sur la dimension de l’ensemble des points base du fibré déterminant sur  $SU_C(r)$ , *Ann. Inst. Fourier (Grenoble)* **57** (2007), 481–490.
- [16] J.-P. Serre, *Linear representations of finite groups*, Graduate Texts in Mathematics **42**, Springer, New York, 1977.